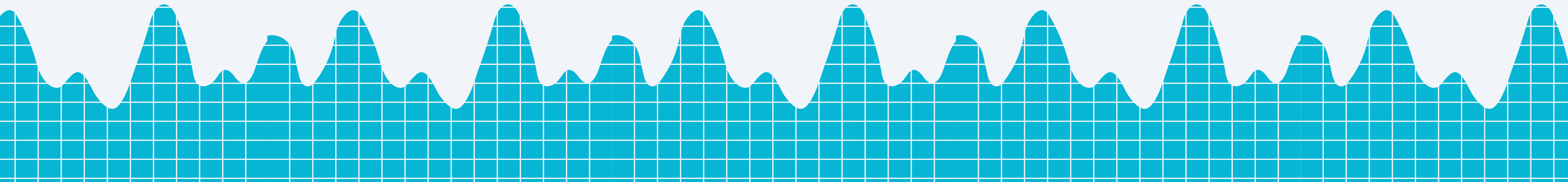


Measuring the Margin: Finding Predictors for Poverty Rates

By Winnie, June, Helen, Akshaj, and Lilly



Goal Of the study

Goal: Analyze key predictors of poverty rates across U.S. states



Poverty is driven by measurable socioeconomic factors, including:

- Education costs
- Per capita personal income
- Population density
- Unemployment rate
- (and other related variables)

Purpose: Build a regression-based model to help the class and team understand what drives impoverished conditions in different communities



Goal Of the study

Scope: US states

Methodology: Three models applied to compare approaches

- Decision Tree
- Random Forest
- Neural Network

Outcome: Draw data-driven conclusions on the strongest predictors of poverty



Data

In our study we used the following predictors for our models:

Education Level

Minimum wage/median wage

Household structure

Crime rate

Substance abuse

Race Ratio

Public transportation

Urbanization Rate

Immigration Rate

Homeless rate

Population density

Unemployment rate

Avg Rent

Avg Annual Income

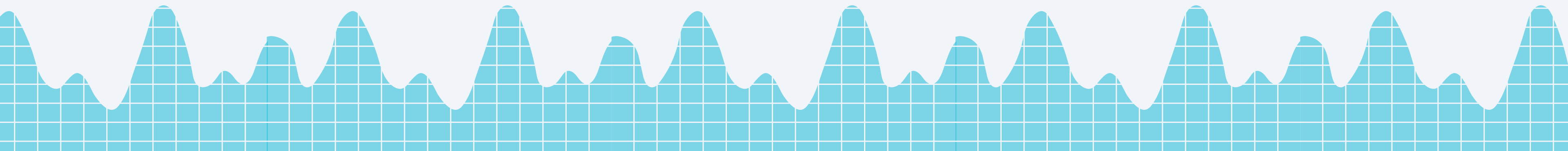
Avg hourly wage

Avg apartment size

Housing price

Avg living expense

Tax



Findings



Education, income, transportation costs, wages, and housing-related indicators consistently ranked as the top predictors across results

- High school graduation was the #1 bivariate predictor ($r = -0.732$) and the #1 Random Forest feature
- The transportation cost index explained 21.5% of bivariate poverty variation ($r = -0.464$; $r^2 = 0.215$)
- Logged income was negatively associated with poverty ($r = -0.370$), explaining 13.7% of bivariate variation



Findings

States with higher completion of a bachelor's degree exhibited significantly lower poverty rates.

- Bachelor's degree completion correlated at $r = -0.406$ with poverty
- In a simple bivariate OLS model, each 1 percentage-point increase in bachelor's degree attainment was associated with a 0.139 percentage-point decrease in poverty; in the full multivariable OLS model, the bachelor coefficient was not significant after controlling for high school graduation and correlated predictors



Findings



Tax rates and minimum wage showed weaker predictive power than anticipated

- Combined tax rate ranked 11th out of 30 in Random Forest importance and 5th by absolute bivariate correlation with poverty ($r = 0.416$)
- Minimum wage coefficient was not significant in the full OLS model ($\beta = -0.198$, $p = 0.062$)
- Tax rate was not significant in the full OLS model ($\beta = 6.775$, $p = 0.277$), suggesting the bivariate tax-poverty pattern is partly explained by other state characteristics



Findings

Crime and drug use showed positive correlations with poverty, but causality is unclear

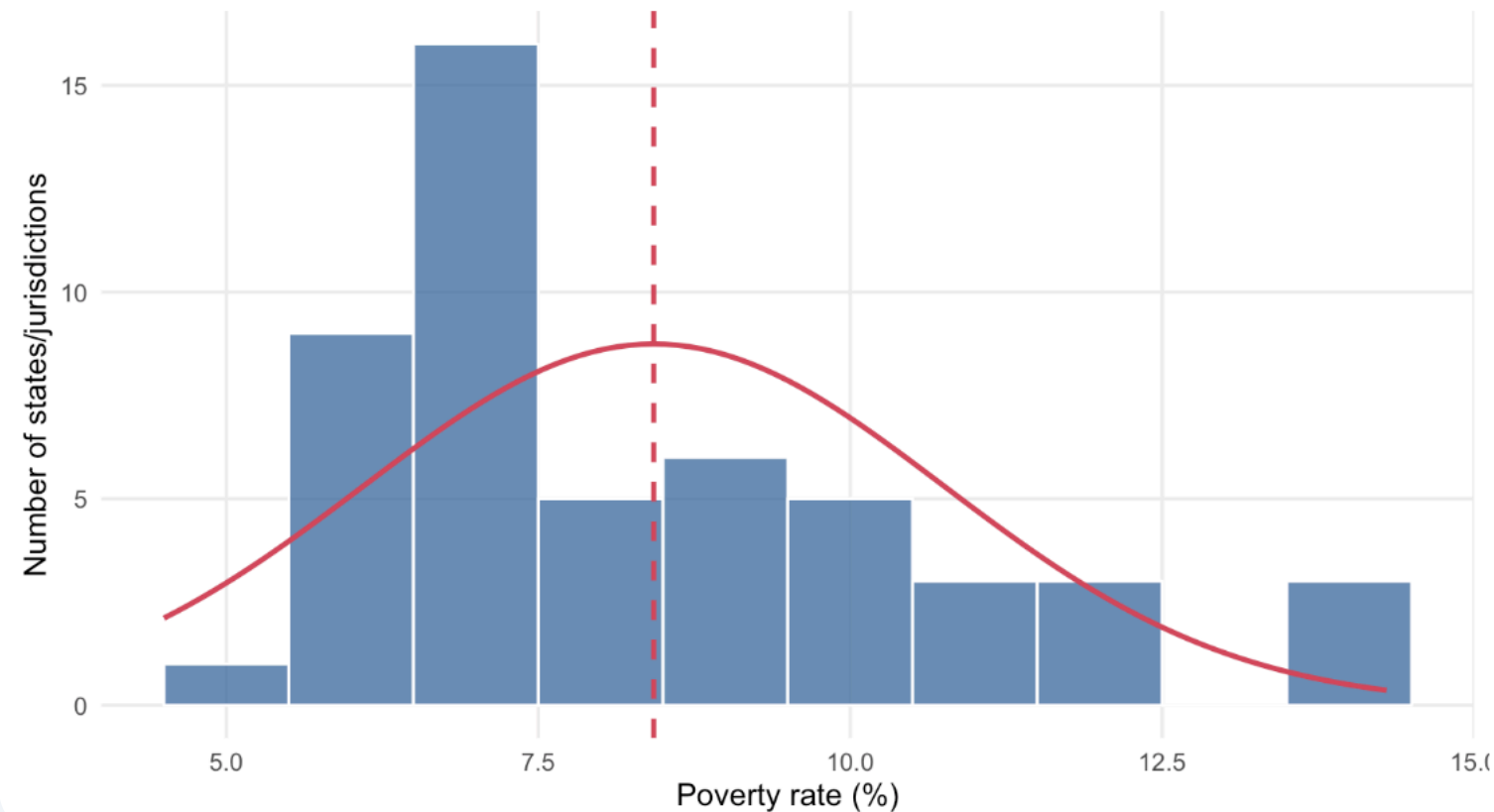
- Crime rate: $r = 0.176$ with poverty
- Drug use: $r = 0.395$ with poverty
- Crime ranked 3rd in the Random Forest importance run, while drug use was significant in full OLS and selected by LASSO but ranked lower in Random Forest importance



EDA

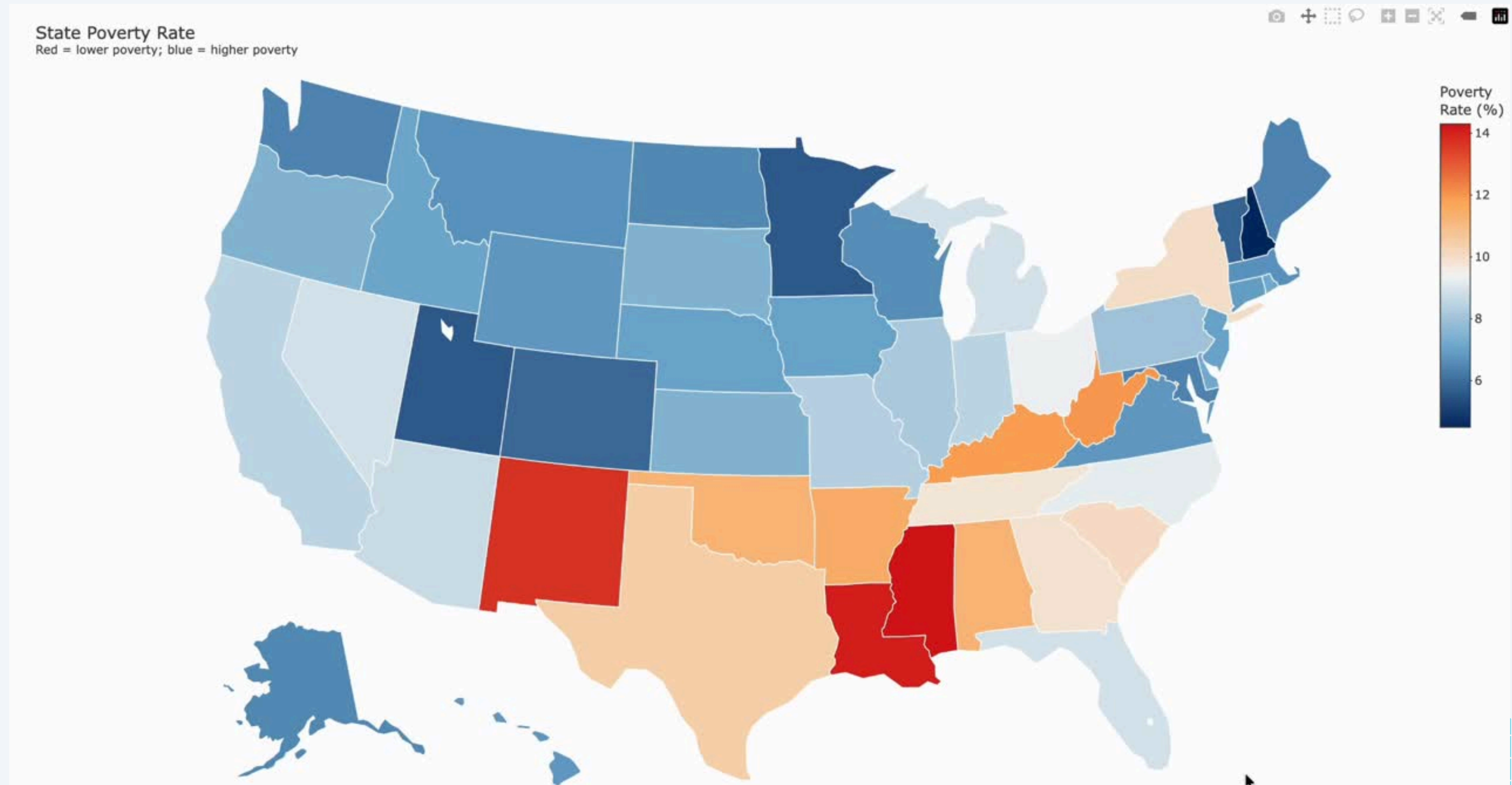
Distribution of Poverty Rates Across States

Slightly right-skewed; dashed line marks the mean

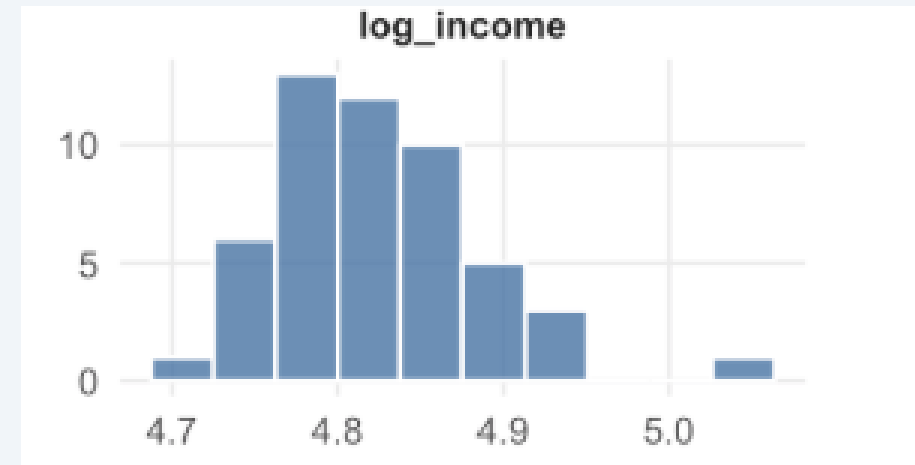
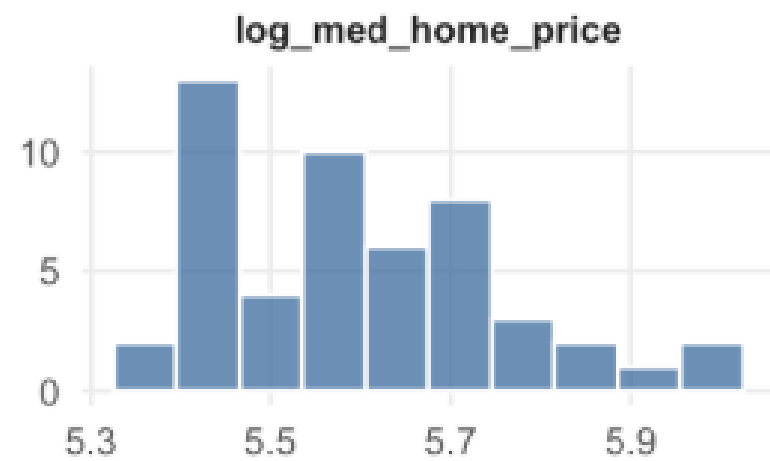


- Highest poverty states: Mississippi (14.3%), Louisiana (14.1%), New Mexico (13.7%), West Virginia (12.0%), and District of Columbia (11.9%)
- Lowest poverty states: New Hampshire (4.5%), Minnesota (5.6%), Utah (5.6%), Vermont (5.8%), and Colorado (5.9%)
- Regional pattern: the South had the highest average poverty rate (10.42%), while the Northeast had the lowest average (6.94%)

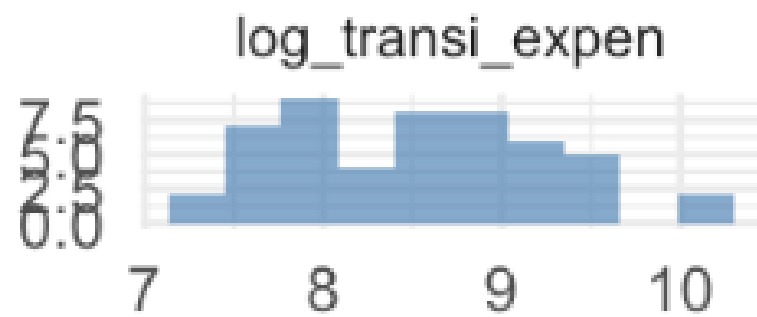
EDA



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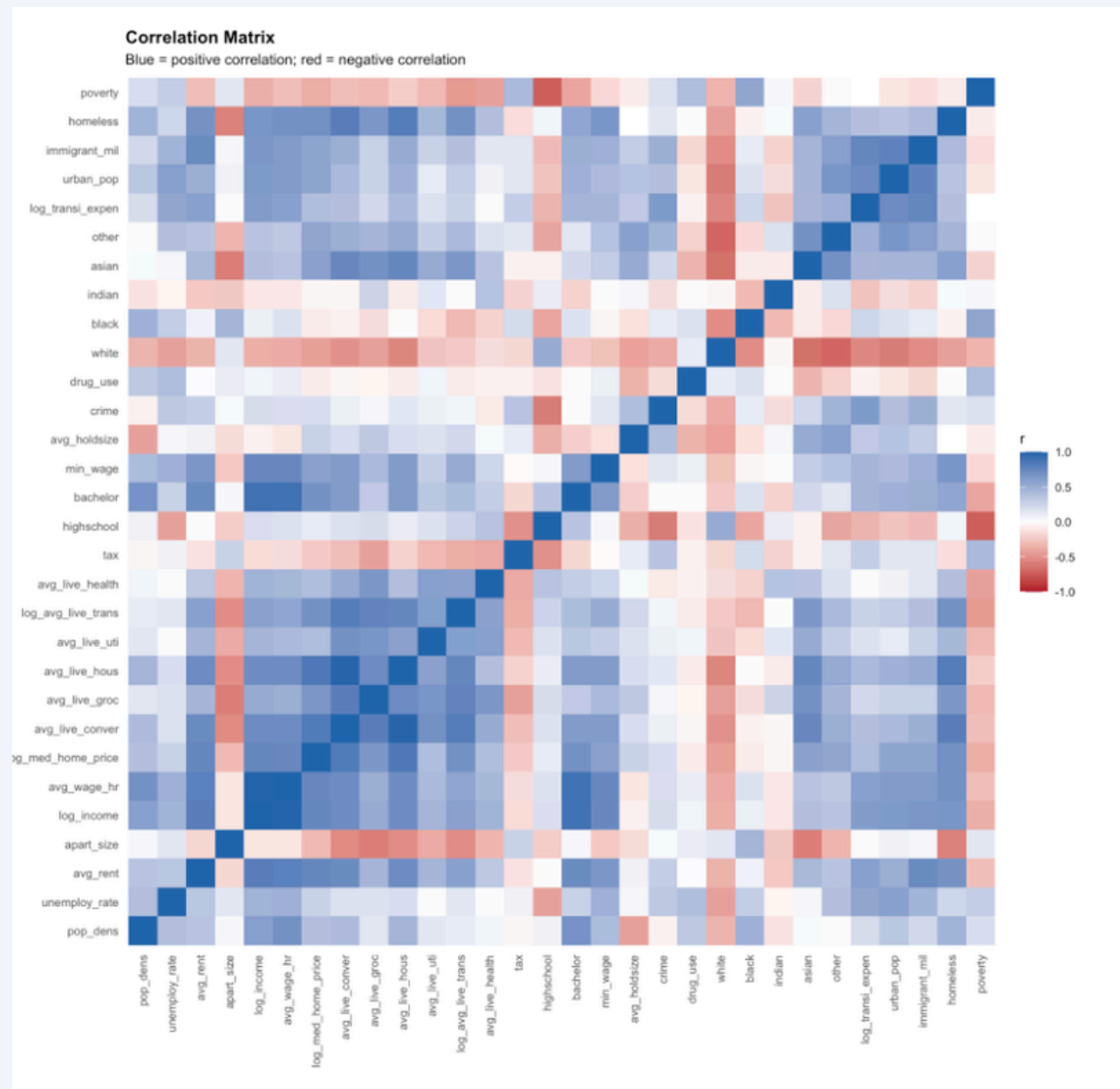
Several predictor variables exhibited strong right-skew behavior, causing us to initiate log transformations:



- `log\_income`: Per capita income (original range: \$50,816 - \$110,917)
- `log\_med\_home\_price`: Median home price (original range: \$229,087 - \$977,237)
- `log\_avg\_live\_trans`: Transportation cost index
- `log\_transi\_expen`: Public transit expenditure

EDA

Correlation Matrix Heat Map



Top Positive Correlations with Poverty:

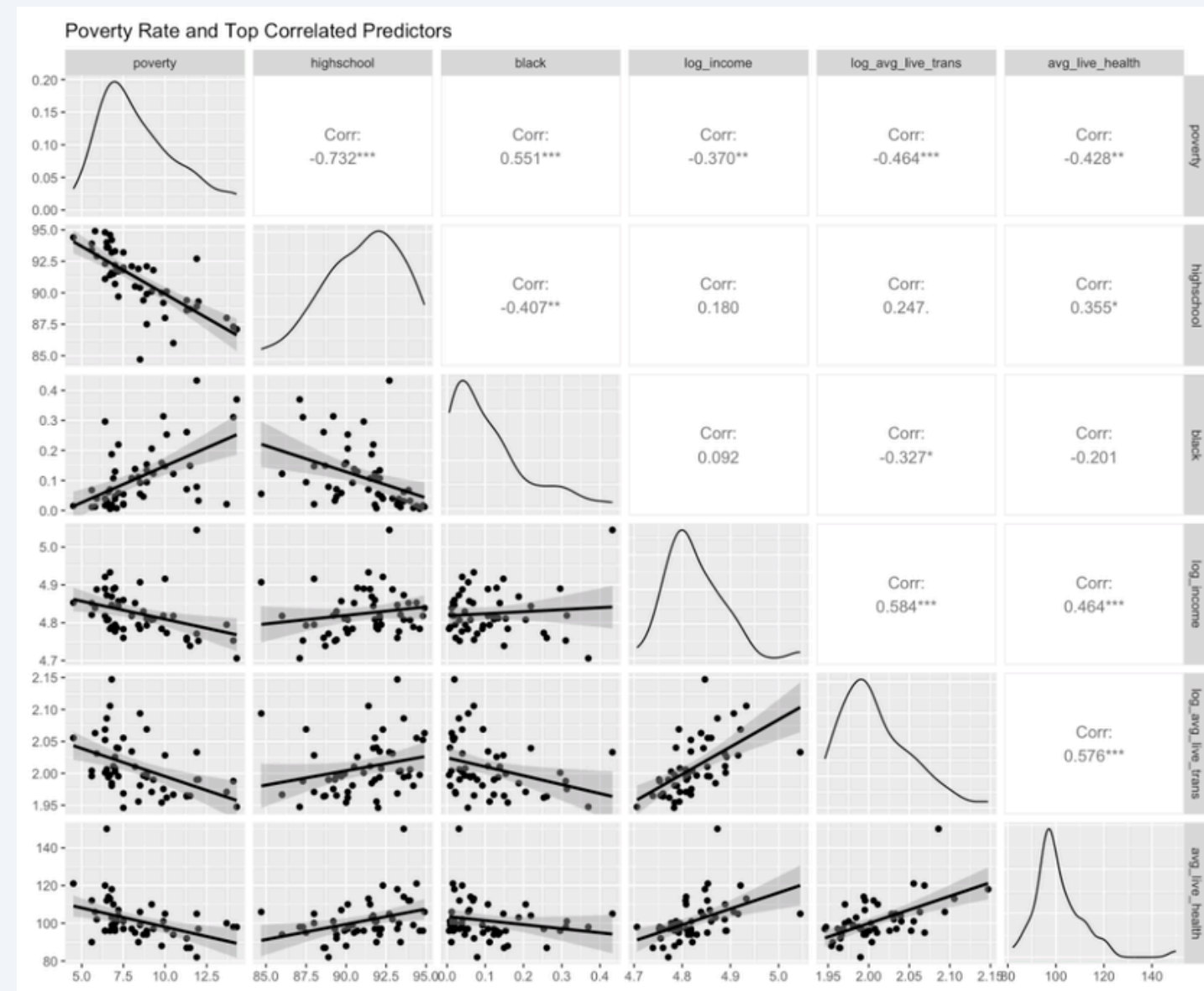
- `tax`: $r = 0.416$
- `drug\_use`: $r = 0.395$

Strong/Moderate Negative Correlations with Poverty:

- `highschool`: $r = -0.732$
- `log\_avg\_live\_trans`: $r = -0.464$
- `avg\_live\_health`: $r = -0.428$

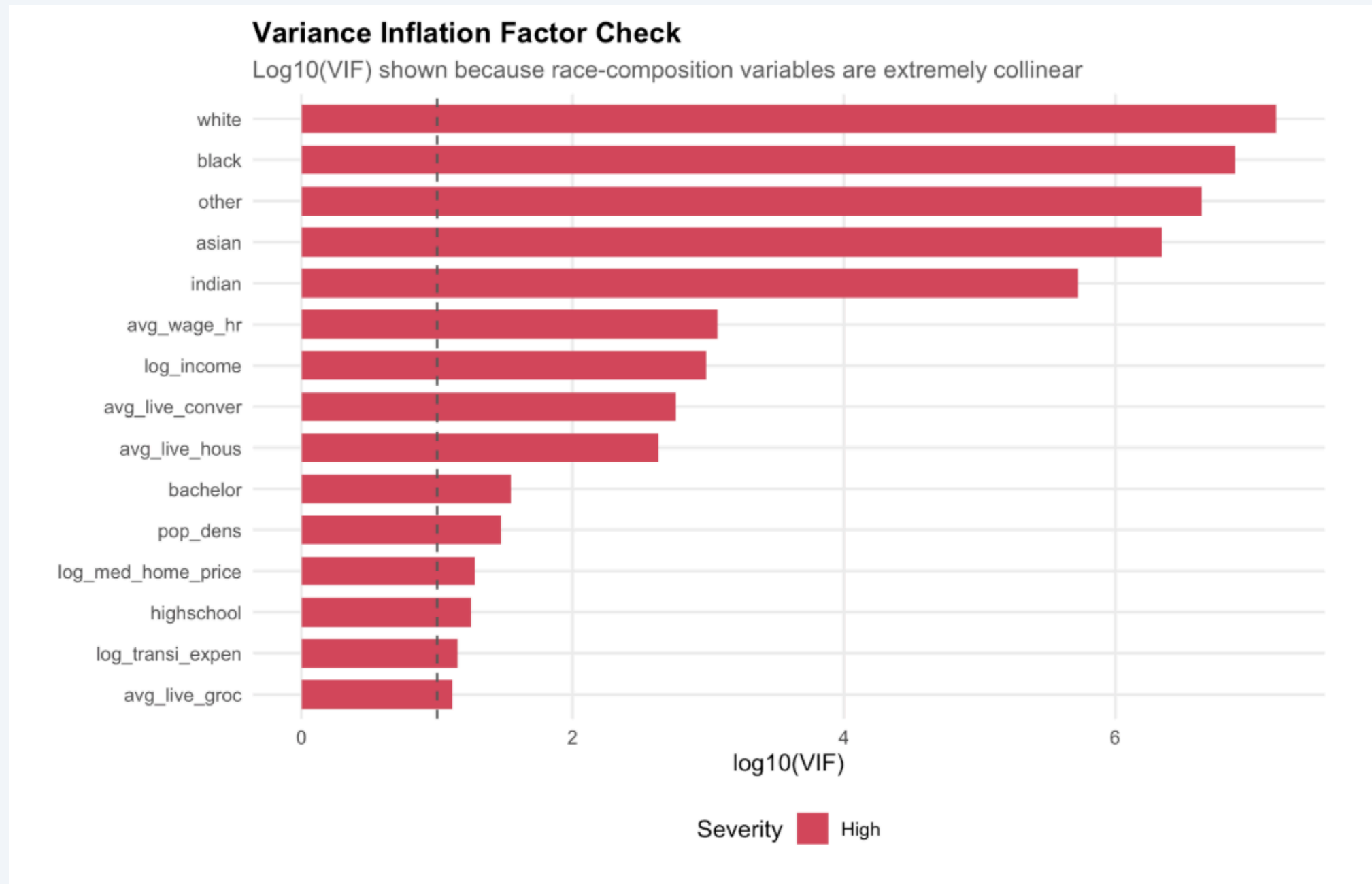
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Poverty Rate and Top Correlated Predictors



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Variance Inflation Factor



Models

01

**Ordinary Least Squares
(OLS) Regression**

02

LASSO Regularization

03

Decision Tree

04

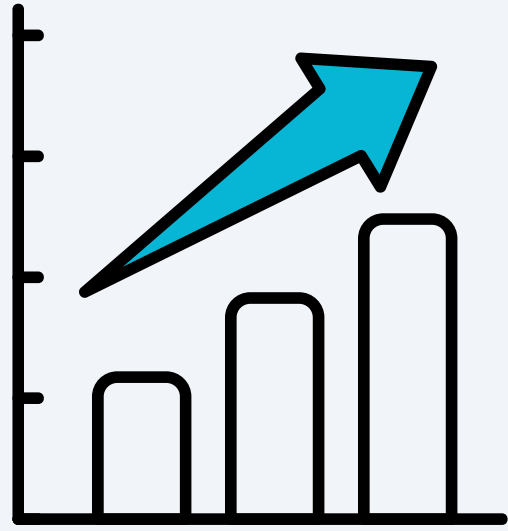
Random Forest

05

**Support Vector
Regression (SVR)**

06

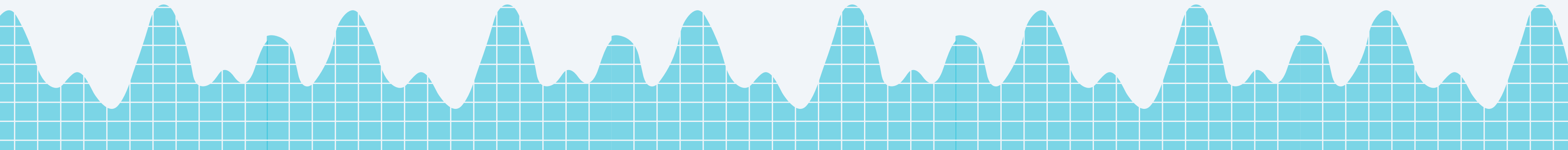
**Deep Neural
Network**



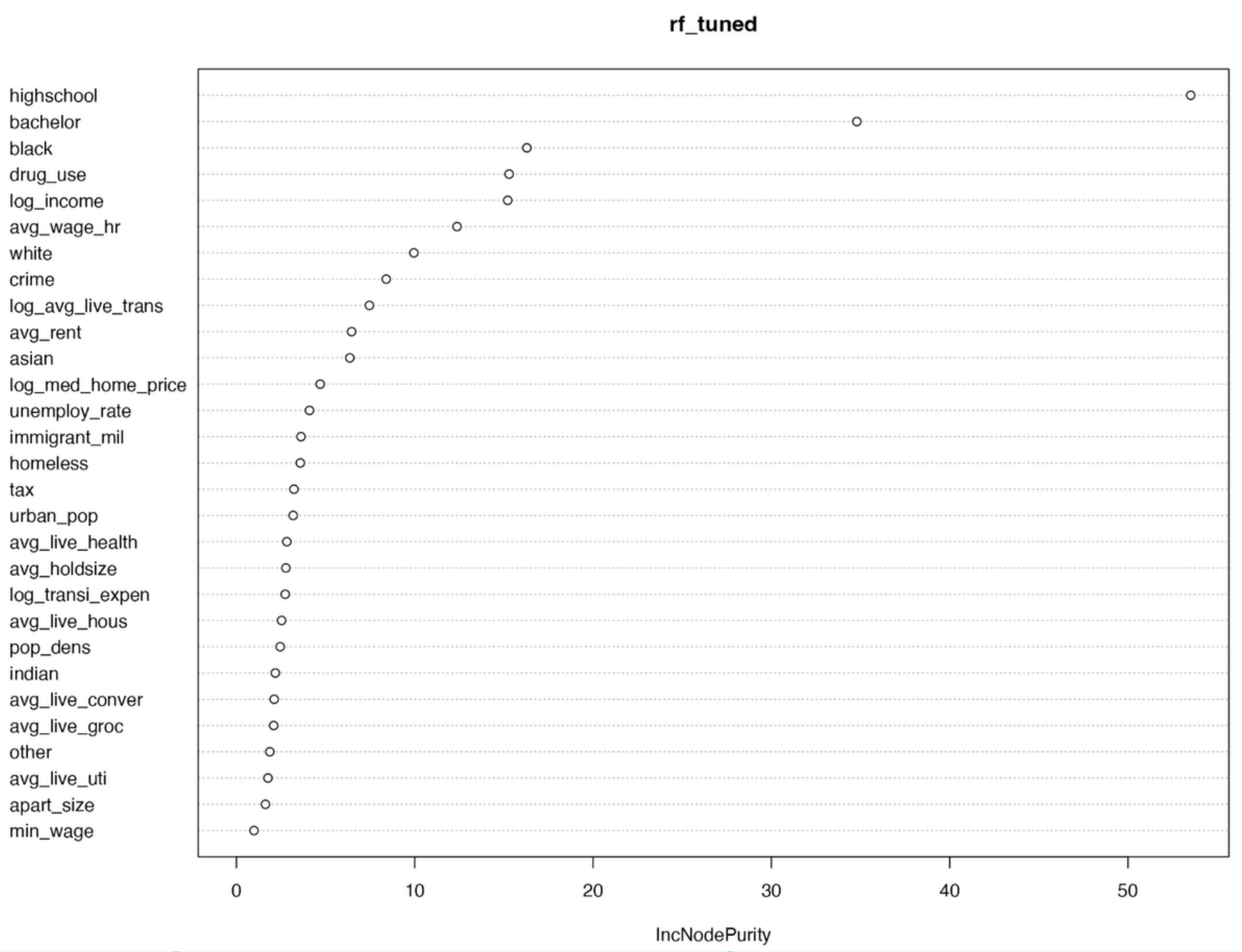
Models

Variance Inflation Factor

Models	MAE (Test)	RSS (Test)	RMSE(Test)	R^2 (Test)	Train RMSE	Train R^2	Overfitting?
OLS Regression	0.771	8.34	1.02	0.89	0.307	0.983	Yes
LASSO	0.593	4.92	0.784	0.918	0.383	0.974	Mild
Decision Tree	1.21	20.6	1.61	0.418	0.867	0.866	Yes
Random Forest	0.319	1.31	0.405	0.964	0.641	0.964	No
SVR	0.506	2.97	0.609	0.985	0.517	0.968	No
Deep Neural Network	0.955	13.5	1.08	0.959	1.124	0.848	No



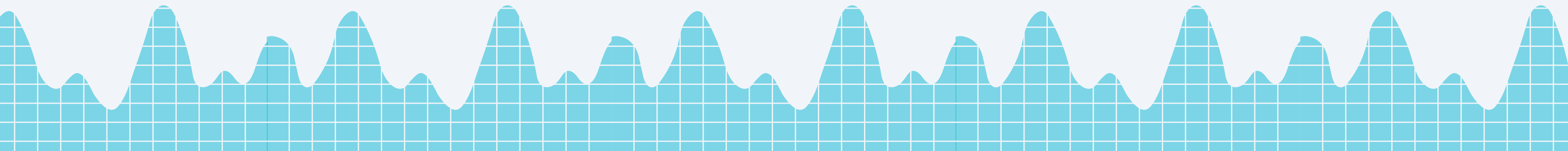
Models



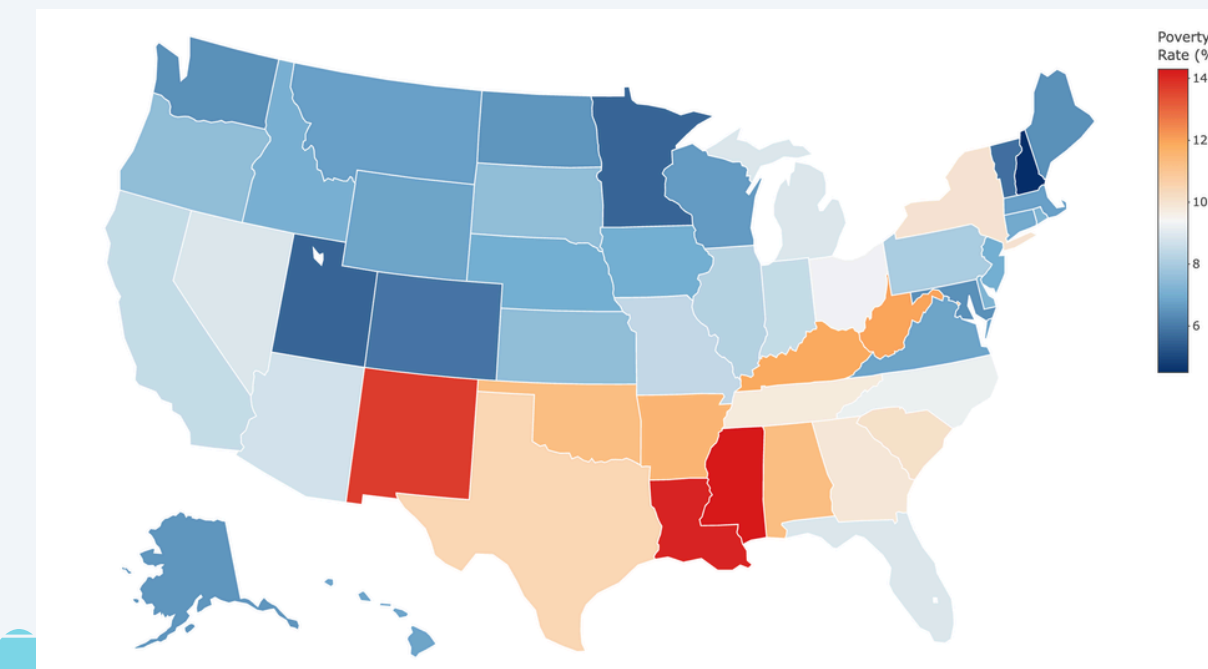
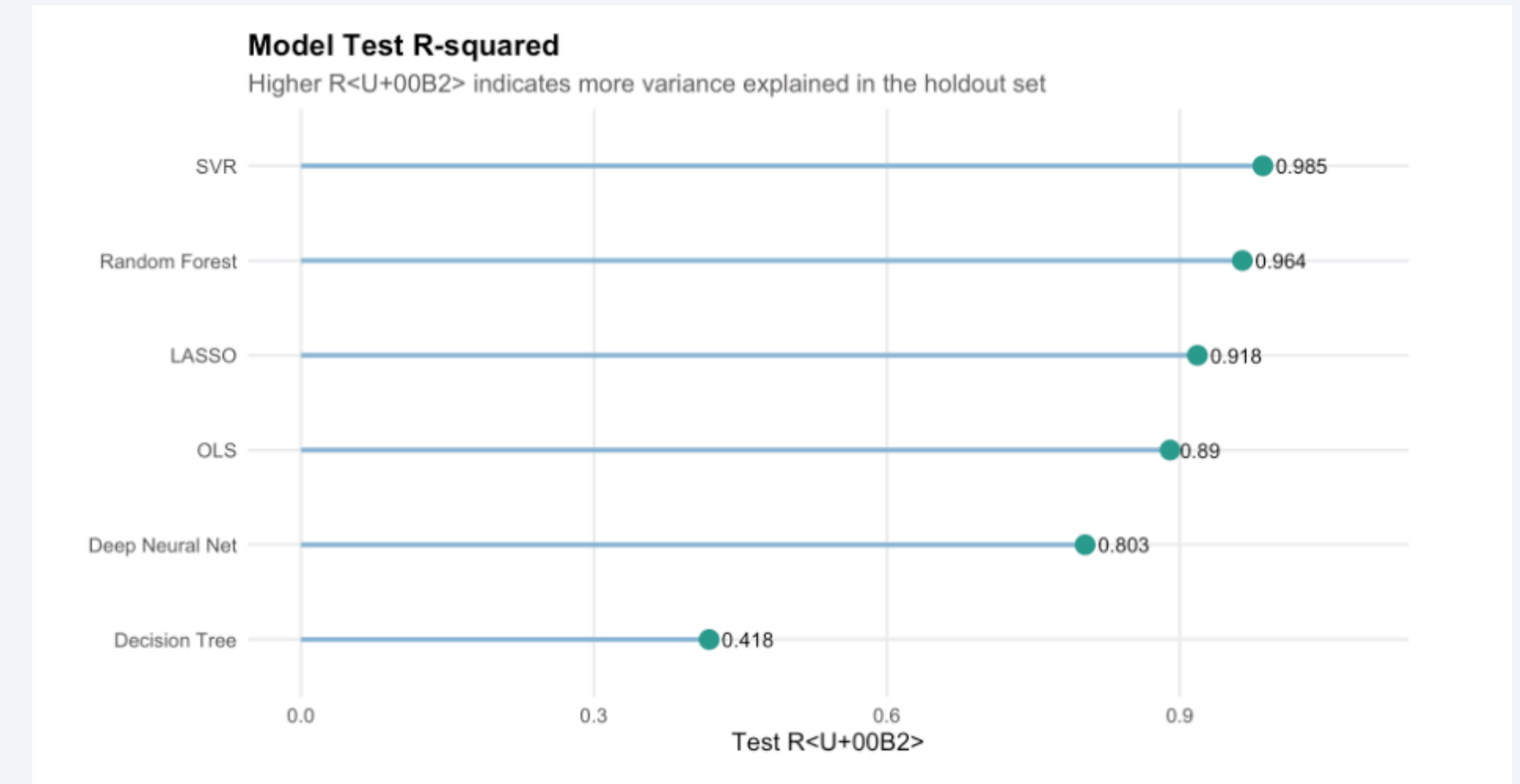
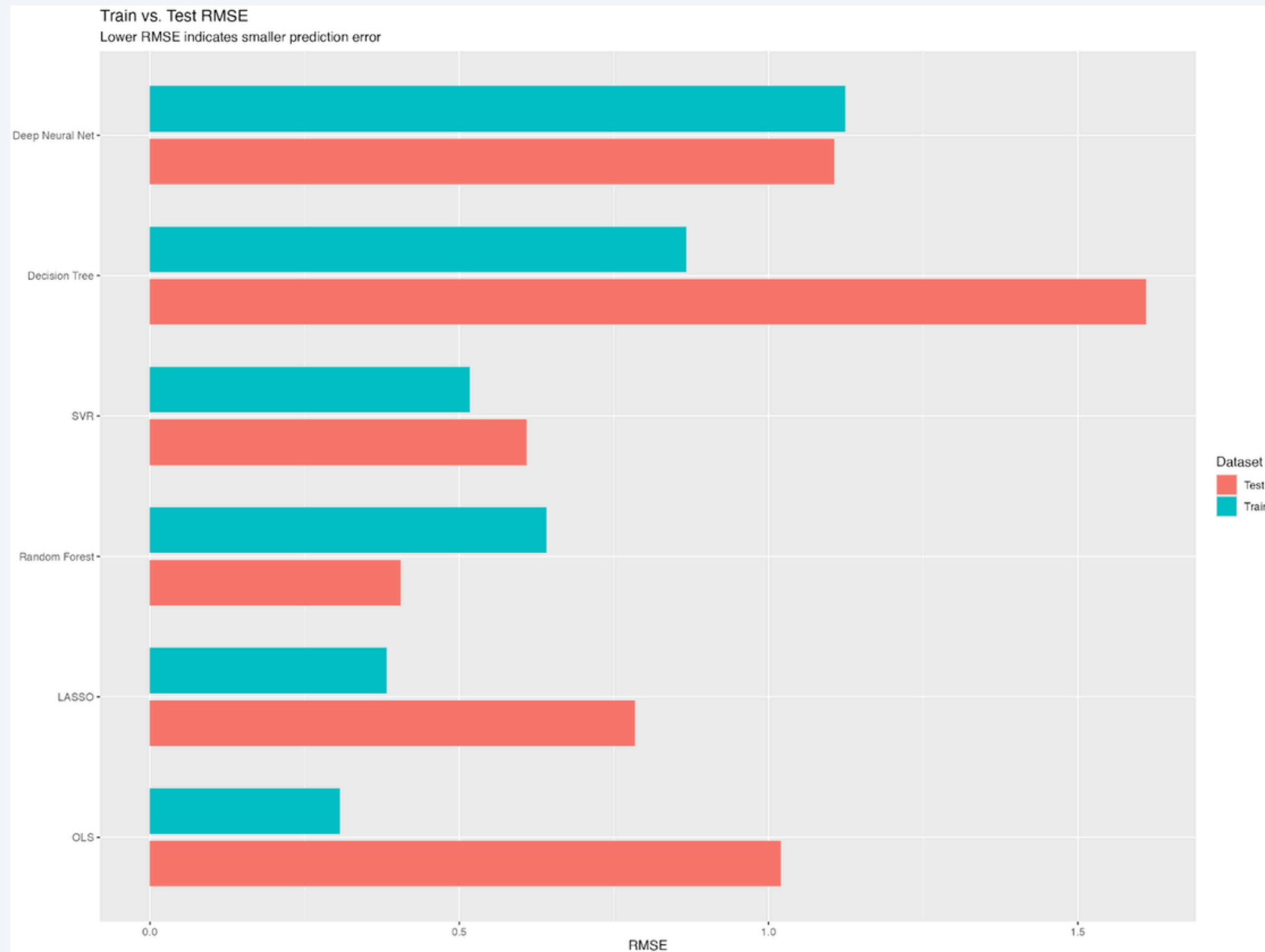
Rank	Variable	Importance Score	Catagory
1	Highschool	15.824	Education
2	bachelor	8.222	Education
3	crime	5.878	Social
4	avg_wage_hr	5.567	Economic
5	log_income	5.248	Economic
6	log_avg_live_trans	4.564	Economic
7	white	4.283	Demographic
8	black	4.6	Demographic
9	asian	3.522	Demographic
10	avg_rent	2.838	Economic

Conclusion/Evaluation

- LASSO results: strongest predictors of state poverty rate are:
 - High school graduation rate
 - Black population share
- Linear model results (direction of association):
 - Higher high school graduation rate → associated with lower poverty
 - Higher Race population share → associated with higher poverty
- Important caveats:
 - These variables may reflect underlying bias in the data
 - Findings could be linked to conclusions unrelated to poverty itself
 - Included because the model was explicitly designed to account for race and education
- Interpretation limits:
 - These are state-level associations only
 - Not causal effects
 - Should not be generalized or interpreted beyond the context of this study



AI Usage



Citations

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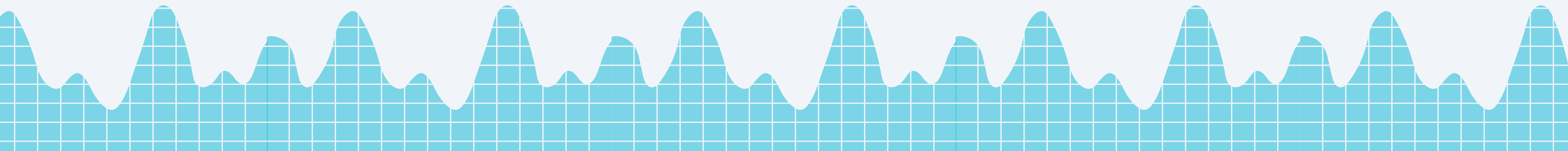
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Thank You!

